

GlobalWafers Co Ltd 6488 ★ 11 May 2026 21:39, UTC

Last Price 785.00 TWD 12 May 2026	Fair Value Estimate 560.00 TWD 12 May 2026 10:02, UTC	Price/FVE 1.40	Market Cap — TWD	Economic Moat™ Narrow	Equity Style Box Large Blend	Uncertainty High	Capital Allocation Standard	ESG Risk Rating Assessment¹ 10 May 2026 05:00, UTC
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Phelix Lee

Senior Equity Analyst

Morningstar

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Important Disclosure

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The primary analyst covering this company does not own its stock.

Reporting Currency: TWD | Trading Currency: TWD
Currency amounts expressed with "\$" are in U.S. dollars (USD) unless otherwise denoted.

The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

GlobalWafers and SAS See Clear Non-Artificial Intelligence Upcycle; SAS Is Our Pick

Analyst Note Phelix Lee, Senior Equity Analyst, 12 May 2026

Share prices of Sino-American Silicon and its subsidiary GlobalWafers surged after earnings, amid upbeat management comments on the 2026 and 2027 outlook. These echo industry commentary on recovery in non-artificial intelligence demand, adding to the vibrant AI demand.

Why it matters: Management now believes demand is shifting from AI-driven to overall strength. The recovery is swifter than we expected and can be attributed to strong analog and power chip demand from faster electrification in non-China vehicles and a better factory automation outlook.

- We view analog and power strength as a tide that lifts all boats. While Japanese peers have an edge in cutting-edge logic and memory, GlobalWafers and other wafer producers benefit from analog, power, and optical chips, as most of them are fabricated on variations of decades-old technology.
- Advanced Wireless Semiconductor Company, another SAS subsidiary, is a gallium arsenide foundry. Apart from AI, it benefits from satellite communications, Wi-Fi 7 power amplifiers for drones, and light ranging products for the auto and industrials sectors.

The bottom line: We lift our fair value estimate on narrow-moats GlobalWafers by 12% to TWD 560; and SAS by 11% to TWD 235 due to stronger-than-expected non-AI recovery. GlobalWafers is expensive to us as we still don't view a structural shortage of wafers, given the capacity roadmap by Chinese peers.

- SAS is our pick as it trades at a mid-50% holding discount to its listed subsidiaries, versus high-40s two months ago. We think this discount is unwarranted as the subsidiaries are consistently profitable, and it reflects

Financial Summary and Key Statistics

	Actual		Forecast	
	2024	2025	2026	2027
Revenue (TWD Mil)	62,626	60,598	67,335	83,368
Revenue Growth %	-11.4	-3.2	11.1	23.8
Operating Income (Mil)	14,118	8,636	12,029	22,418
Operating Margin %	22.5	14.2	17.9	26.9
Adjusted EBITDA (Mil)	18,010	17,829	24,730	36,260
Adjusted EBITDA Margin %	28.8	29.4	36.7	43.5
Earnings Per Share (Diluted) (TWD)	20.52	16.27	19.62	37.24
Adjusted Earnings Per Share (Diluted) (TWD)	20.52	16.27	19.62	37.24
Adjusted EPS Growth %	-53.1	-20.7	20.6	89.8
Price/Earnings	18.6	25.0	42.6	22.4
Price/Book	2.0	2.1	4.1	3.6
EV/EBITDA	10.5	12.5	16.6	11.3
Free Cash Flow Yield %	—	—	—	—

Source: Morningstar Valuation Model. Data as of 12 May 2026.

GlobalWafers Co Ltd

6488★

11 May 2026 21:39, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
785.00 TWD	560.00 TWD	1.40	— TWD	Narrow	Large Blend	High	Standard	10 May 2026 05:00, UTC
12 May 2026	12 May 2026 10:02, UTC							

Sector	Industry
Technology	Semiconductors

Business Description

GlobalWafers is 46.64% owned by Sino-American Silicon Products after being spun off in 2011. The main source of the company's revenue is selling silicon wafers to chipmakers like TSMC and Texas Instruments. It also derives sales from gallium nitride products and green energy. GlobalWafers runs 18 production sites in nine countries across Asia, North America, and Europe. GlobalWafers is headquartered in Hsinchu, Taiwan, and employs over 7,300 people. GlobalWafers had nearly 17% market share in silicon wafers in 2024.

the market's preference for pure-play names with clearer themes.

► We think better disclosure and taking the majority share in subsidiaries may narrow the discount. Enhanced disclosure may mean joint earnings calls or investor days involving SAS and listed units. Taking the majority share is less likely in the near term as GlobalWafers needs capital to expand.

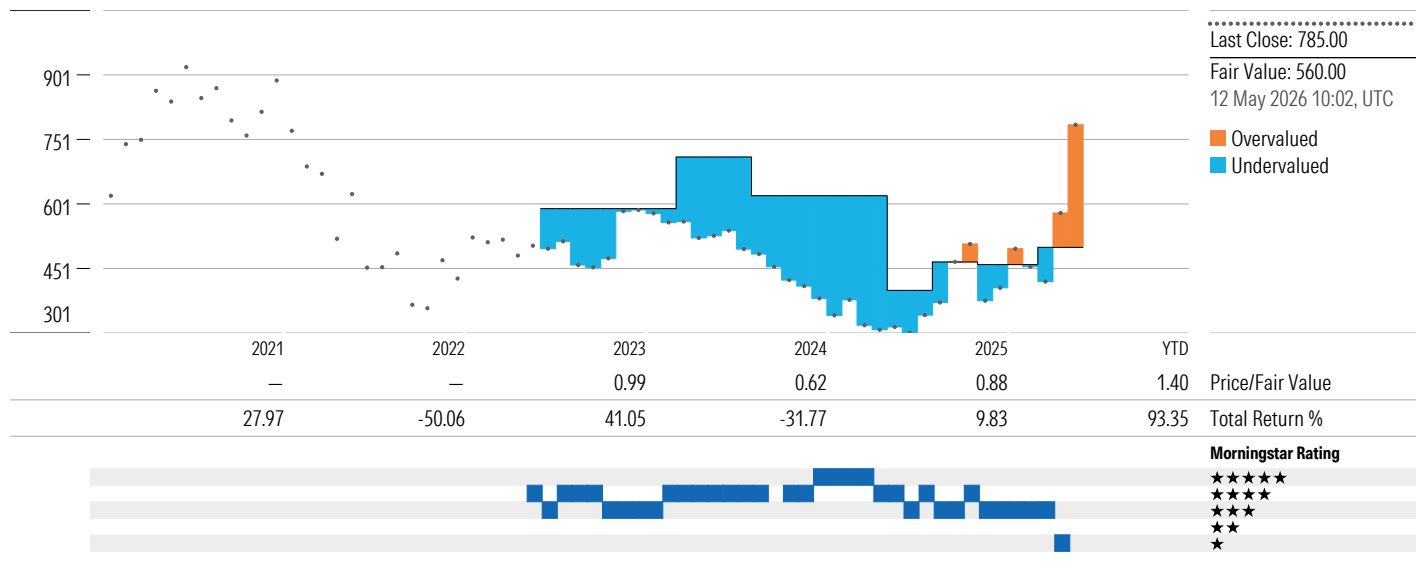
GlobalWafers' March-quarter revenue was down 4% sequentially to TWD 14 billion, but operating margin sank 586 basis points to 10.5%. The lower profitability stems from ramp-ups in Japan and the US, higher-than-usual energy costs amid blizzards in the US, plus the lack of one-off subsidy boosts. We believe the quarter's performance is the lowest point of the year, as spot prices were recovering around the end of the quarter.

SAS' March-quarter revenue was flat sequentially at TWD 19.4 billion, and operating margin plunged 537 basis points to 11.9%. SAS fared better than GlobalWafers because its other subsidiaries AWSC, and Taiwan Specialty Materials, outpaced the group average. The two subsidiaries enjoyed 27.4% and 40.2% operating margins, respectively. AWSC, in particular, benefits from resilient high-end smartphone power amplifier demand, and like its peer Win Semiconductors, has a footing in optical products for data centers, such as copackaged optics.

Part of our fair value estimate increase stems from a long near-term recovery, as we extend the upswing into 2028, followed by a more moderate 2029. However, we do not buy the structural shortage argument for three reasons. First, vehicle electrification is not a new phenomenon. Currently, the level of semiconductor content in vehicles supports a higher legal autonomous driving classification than what the vehicles are sold at, due to future-proofing and in anticipation of regulatory progress. Second, analog, power, and sensor semiconductors are made on legacy nodes that are easier for Chinese entrants to take share; and, lastly, China's push for self-sufficiency. ■■■

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Price vs. Fair Value



Competitors

	GlobalWafers Co Ltd 6488	Soitec SA Shar... reverse split SOI	National Silic...Co Ltd Class A 688126	SUMCO Corp 3436
	Last Close 785.00 Fair Value 560.00 Uncertainty: High	Last Close 161.70 Fair Value 42.00 Uncertainty: Very High	Last Close 24.19 Fair Value 6.60 Uncertainty: High	Last Close 3,760.00 Fair Value 1,200.00 Uncertainty: High
Economic Moat	Narrow	Narrow	None	None
Currency	TWD	EUR	CNY	JPY
Fair Value	560.00 12 May 2026 10:02, UTC	42.00 11 May 2026 22:55, UTC	6.60 11 May 2026 16:57, UTC	1,200.00 11 May 2026 16:59, UTC
1-Star Price	868.00	73.50	10.23	1,860.00
5-Star Price	336.00	21.00	3.96	720.00
Assessment	Overvalued 12 May 2026	Overvalued 11 May 2026	Overvalued 12 May 2026	Overvalued 12 May 2026
Morningstar Rating	★11 May 2026 05:00, UTC	★11 May 2026 05:00, UTC	★11 May 2026 05:00, UTC	★11 May 2026 05:00, UTC
Analyst	Phelix Lee, Senior Equity Analyst	Javier Correoneo, Senior Equity Analyst	Phelix Lee, Senior Equity Analyst	Phelix Lee, Senior Equity Analyst
Capital Allocation	Standard	Standard	Poor	Standard
Price/Fair Value	1.40	3.85	3.67	3.01
Price/Sales	6.60	7.39	17.78	3.08
Price/Book	4.28	3.92	4.75	2.18
Price/Earnings	54.72	573.77	—	—
Dividend Yield	1.20%	0.00%	0.00%	0.79%
Market Cap	375.32 Bil	5.78 Bil	79.95 Bil	1,314.92 Bil
52-Week Range	278.50—849.00	22.62—180.00	16.61—29.97	937.00—3,765.00
Investment Style	Large Blend	Mid Blend	Large Blend	Mid Value

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Morningstar Valuation Model Summary

Financials as of 12 May 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (TWD Mil)	70,652	62,626	60,598	67,335	83,368	87,363	91,806	100,079
Operating Income (TWD Mil)	20,059	14,118	8,636	12,029	22,418	28,438	21,757	29,331
EBITDA (TWD Mil)	30,630	18,010	17,829	24,730	36,260	42,219	35,125	42,335
Adjusted EBITDA (TWD Mil)	30,630	18,010	17,829	24,730	36,260	42,219	35,125	42,335
Net Income (TWD Mil)	19,772	9,846	7,800	9,405	17,850	22,909	17,800	24,092
Adjusted Net Income (TWD Mil)	19,772	9,846	7,800	9,405	17,850	22,909	17,800	24,092
Free Cash Flow To The Firm (TWD Mil)	42,642	-27,059	-32,416	16,552	11,158	16,766	18,726	16,231
Weighted Average Diluted Shares Outstanding (Mil)	452	480	479	479	479	479	479	479
Earnings Per Share (Diluted) (TWD)	43.73	20.52	16.27	19.62	37.24	47.80	37.14	50.27
Adjusted Earnings Per Share (Diluted) (TWD)	43.73	20.52	16.27	19.62	37.24	47.80	37.14	50.27
Dividends Per Share (TWD)	19.00	19.00	11.00	9.70	11.70	22.30	28.60	22.20

Margins & Returns as of 12 May 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	21.5	28.4	22.5	14.2	17.9	26.9	32.6	23.7	29.3	26.4
EBITDA Margin %	—	43.4	28.8	29.4	36.7	43.5	48.3	38.3	42.3	—
Adjusted EBITDA Margin %	—	43.4	28.8	29.4	36.7	43.5	48.3	38.3	42.3	41.8
Net Margin %	18.9	28.0	15.7	12.9	14.0	21.4	26.2	19.4	24.1	21.0
Adjusted Net Margin %	18.9	28.0	15.7	12.9	14.0	21.4	26.2	19.4	24.1	21.0
Free Cash Flow To The Firm Margin %	-12.1	60.4	-43.2	-53.5	24.6	13.4	19.2	20.4	16.2	18.8

Growth & Ratios as of 12 May 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	-4.8	0.5	-11.4	-3.2	11.1	23.8	4.8	5.1	9.0	10.5
Operating Income Growth %	-29.8	-19.7	-29.6	-38.8	39.3	86.4	26.9	-23.5	34.8	27.7
EBITDA Growth %	-7.4	20.0	-41.2	-1.0	38.7	46.6	16.4	-16.8	20.5	21.1
Adjusted EBITDA Growth %	-11.3	20.0	-41.2	-1.0	38.7	46.6	16.4	-16.8	20.5	18.9
Earnings Per Share Growth %	-21.1	31.8	-53.1	-20.7	20.6	89.8	28.3	-22.3	35.4	25.3
Adjusted Earnings Per Share Growth %	-21.1	31.8	-53.1	-20.7	20.6	89.8	28.3	-22.3	35.4	25.3

Valuation as of 12 May 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earnings	13.4	18.6	25.0	42.6	22.4	17.5	22.5	16.6
Price/Sales	3.6	2.9	3.2	5.9	4.8	4.6	4.3	4.0
Price/Book	4.0	2.0	2.1	4.1	3.6	3.3	3.2	2.9
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	7.6	10.5	12.5	16.6	11.3	9.7	11.7	9.7
EV/EBIT	11.5	13.4	25.7	34.1	18.3	14.4	18.8	14.0
Dividend Yield %	3.2	5.0	2.7	1.2	1.4	2.7	3.4	2.7
Dividend Payout %	43.4	92.6	67.6	49.4	31.4	46.7	77.0	44.2
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 12 May 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	10.5	4.4	3.6	4.3	7.9	9.8	7.6	9.9
ROE %	29.8	10.8	8.4	9.6	16.2	18.7	14.1	17.2
ROIC %	26.2	11.1	5.5	28.2	55.5	68.1	57.5	83.1

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Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year , ends 31 Dec	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	17.6	26.7	27.7	19.8	17.5	15.6	13.8	12.0
Assets/Equity	2.8	2.5	2.3	2.2	2.1	1.9	1.9	1.7
Net Debt/EBITDA	-0.5	0.6	0.6	1.9	1.1	0.6	0.5	0.1
Total Debt/EBITDA	1.8	3.7	4.2	2.8	1.8	1.4	1.5	1.2
EBITDA/ Net Interest Expense	-11.8	-7.2	-15.9	43.2	89.4	415.6	-181.9	-87.4

Forecast Revisions as of 12 May 2026	2026		2027		2028	
Prior data as of 13 Mar 2026	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	560.00	215.68	—	—	—	—
Revenue (TWD Mil)	67,335	93,939	83,368	111,312	87,363	108,267
Operating Income (TWD Mil)	12,029	16,324	22,418	28,928	28,438	18,694
EBITDA (TWD Mil)	24,730	32,575	36,260	44,980	42,219	34,632
Net Income (TWD Mil)	9,405	6,979	17,850	12,330	22,909	8,180
Earnings Per Share (Diluted) (TWD)	19.62	11.81	37.24	20.87	47.80	13.84
Adjusted Earnings Per Share (Diluted) (TWD)	19.62	11.81	37.24	20.87	47.80	13.84
Dividends Per Share (TWD)	9.70	8.00	11.70	7.60	22.30	13.50

Key Valuation Drivers as of 12 May 2026

Cost of Equity %	9.9
Pre-Tax Cost of Debt %	5.6
Weighted Average Cost of Capital %	9.2
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	7.6
Stage II Investment Rate %	30.2
Perpetuity Year	15

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 12 May 2026

	TWD Mil
Present Value Stage I	61,051
Present Value Stage II	98,525
Present Value Stage III	154,652
Total Firm Value	314,227
Cash and Equivalents	19,484
Debt	74,254
Other Adjustments	0
Equity Value	259,457
Projected Diluted Shares	479
Fair Value per Share (TWD)	560.00

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Income Statement (TWD)

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Year , ends 31 Dec								
Revenue (Mil)	70,652	62,626	60,598	67,335	83,368	87,363	91,806	100,079
Cost of Goods Sold (Mil)	44,211	42,823	45,974	49,063	53,986	51,516	62,251	62,362
Gross Profit (Mil)	26,441	19,804	14,624	18,272	29,382	35,847	29,555	37,716
Selling, General, Administrative & Other Expenses (Mil)	4,002	3,365	3,770	3,892	4,448	4,743	4,972	5,390
Advertising & Marketing Expenses (Mil)	16	3	-1	0	0	0	0	0
Research & Development (Mil)	2,364	2,317	2,218	2,351	2,515	2,666	2,826	2,996
Depreciation & Amortization (if reported separately) (Mil)	0	0	0	0	0	0	0	0
Adjusted Operating Income (Mil)	20,059	14,118	8,636	12,029	22,418	28,438	21,757	29,331
Financial Non-Cash (Gains)/Losses (Mil)	-3,427	4,535	658	0	0	0	0	0
Irregular Cash (Gains)/Losses (Mil)	-412	-357	-902	-300	-300	-300	-300	-300
Operating Income (Mil)	23,897	9,940	8,881	12,329	22,718	28,738	22,057	29,631
Net Interest Expense (Mil)	-2,600	-2,489	-1,124	572	406	102	-193	-485
Income Tax Expense (Mil)	6,727	2,590	2,205	2,351	4,463	5,727	4,450	6,023
After-Tax Items (Mil)	0	0	0	0	0	0	0	0
(Minority Interest) (Mil)	2	7	0	0	0	0	0	0
Net Income (Mil)	19,772	9,846	7,800	9,405	17,850	22,909	17,800	24,092
Adjusted Net Income (Mil)	19,772	9,846	7,800	9,405	17,850	22,909	17,800	24,092
Weighted Average Diluted Shares Outstanding (Mil)	452	480	479	479	479	479	479	479
Diluted Earnings Per Share	43.73	20.52	16.27	19.62	37.24	47.80	37.14	50.27
Diluted Adjusted Earnings Per Share	43.73	20.52	16.27	19.62	37.24	47.80	37.14	50.27
Dividends Per Common Share (TWD)	19.00	19.00	11.00	9.70	11.70	22.30	28.60	22.20
EBITDA (Mil)	30,630	18,010	17,829	24,730	36,260	42,219	35,125	42,335
Adjusted EBITDA (Mil)	30,630	18,010	17,829	24,730	36,260	42,219	35,125	42,335

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Key Cash Flow Items (TWD)

Fiscal Year, ends 31 Dec	Actual			Forecast as of 12 May				
	2023	2024	2025	2026	2027	2028	2029	2030
Cash from Working Capital (Mil)	57,977	400	-17,094	3,288	-4,558	-9,705	-1,987	-10,178
(Capital Expenditures) (Mil)	-36,757	-48,319	-31,938	-9,000	-16,000	-10,000	-10,000	-10,000
Depreciation (Mil)	6,714	8,048	8,927	12,378	13,518	13,456	13,041	12,676
Amortization (Mil)	20	22	22	23	24	25	26	28
Net New (Investment), Organic (Mil)	27,954	-39,850	-40,083	6,689	-7,016	-6,224	1,080	-7,474
(Purchases)/Sales of Companies & Assets (Mil)	282	380	86	0	0	0	0	0
Net New (Investment), Total (Mil)	28,236	-39,471	-39,997	6,689	-7,016	-6,224	1,080	-7,474
Other Non-Cash Items, From Cash Flows (Mil)	0	0	0	0	0	0	0	0
Free Cash Flow to the Firm (Mil)	42,642	-27,059	-32,416	16,552	11,158	16,766	18,726	16,231

Balance Sheet (TWD)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Assets								
Cash and Equivalents (Mil)	26,165	38,929	19,484	21,610	23,873	33,841	35,569	45,355
Inventory (Mil)	9,359	11,238	10,399	11,097	12,211	11,652	14,080	14,105
Accounts Receivable (Mil)	10,016	10,196	10,063	11,182	13,844	14,508	15,245	16,619
Net Property, Plant and Equipment (Mil)	73,181	119,944	108,027	104,649	107,131	103,675	100,634	97,958
Goodwill (Mil)	2,252	2,373	2,251	2,251	2,251	2,251	2,251	2,251
Other Intangibles (Mil)	94	75	68	45	21	-4	-30	-58
Other Operating Assets (Mil)	4,754	9,571	11,085	11,085	11,085	11,085	11,085	11,085
Non-Operating Assets (Mil)	63,167	32,254	56,966	56,966	56,966	56,966	56,966	56,966
Total Assets (Mil)	188,988	224,581	218,343	218,886	227,382	233,975	235,801	244,282
Liabilities								
Accounts Payable (Mil)	14,256	14,992	11,709	12,496	13,750	13,121	15,855	15,883
Debt (Mil)	54,542	65,076	74,254	69,254	64,254	59,254	54,254	49,254
Other Operating Liabilities (Mil)	49,897	49,326	34,509	34,509	34,509	34,509	34,509	34,509
Non-Operating Liabilities (Mil)	3,840	4,160	4,576	4,576	4,576	4,576	4,576	4,576
Total Liabilities (Mil)	122,534	133,553	125,048	120,834	117,088	111,459	109,193	104,222
Equity								
Shareholders' Equity (Mil)	66,450	91,031	93,299	98,055	110,298	122,519	126,612	140,064
Minority Interest (Mil)	4	-3	-3	-3	-3	-3	-3	-3
Total Equity (Mil)	66,454	91,028	93,295	98,051	110,294	122,515	126,608	140,061

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785.00 TWD 12 May 2026	560.00 TWD 12 May 2026 10:02, UTC	1.40	— TWD	Narrow	Large Blend	High	Standard	 10 May 2026 05:00, UTC

ESG Risk Rating Breakdown

Exposure

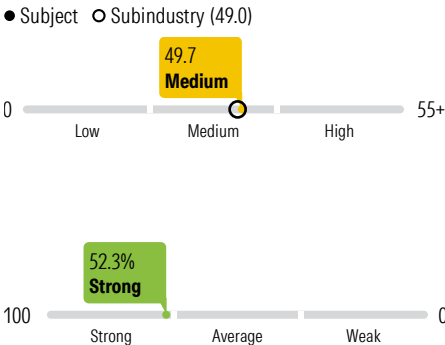
Company Exposure¹	49.7
- Manageable Risk	45.7
Unmanageable Risk²	4.0

Management

Manageable Risk	45.7
- Managed Risk³	23.9
Management Gap⁴	21.8

Overall Unmanaged Risk

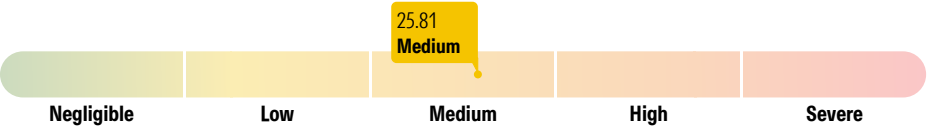
25.8



- ▶ Exposure represents a company's vulnerability to ESG risks driven by their business model
- ▶ Exposure is assessed at the Subindustry level and then specified at the company level
- ▶ Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- ▶ Management measures a company's ability to manage ESG risks through its commitments and actions
- ▶ Management assesses a company's efficiency on ESG programs, practices, and policies
- ▶ Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 52.3% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment⁵



ESG Risk Rating is of May 10, 2026. Highest Controversy Level is as of May 08, 2026. Sustainability Subindustry: Semiconductor Design and Manufacturing. Sustainability provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainability's scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 10 May 2026

Company Name	Exposure	Management	ESG Risk Rating
GlobalWafers Co Ltd	49.7 Medium 0 — 55+	52.3 Strong 100 — 0	25.8 Medium 0 — 40+
Soitec SA	50.4 Medium 0 — 55+	57.7 Strong 100 — 0	23.6 Medium 0 — 40+
SUMCO Corp	50.1 Medium 0 — 55+	44.3 Average 100 — 0	29.7 Medium 0 — 40+
Sino-American Silicon Products Inc	49.0 Medium 0 — 55+	41.2 Average 100 — 0	30.6 High 0 — 40+
National Silicon...try Group Co Ltd	53.3 Medium 0 — 55+	24.4 Weak 100 — 0	41.3 Severe 0 — 40+

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Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our single-point star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a

long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in working capital accounts, and capital spending. Based on these projections, we calculate earnings before interest,

after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

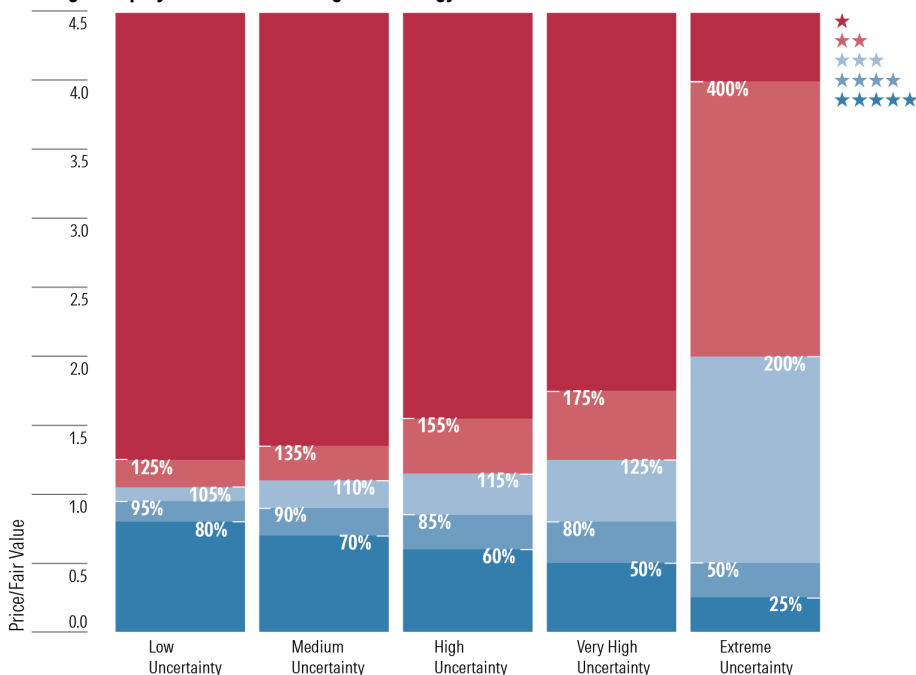
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score.

Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

Ratings should not be used as the sole basis in evaluating a company or security. Ratings involve unknown risks and uncertainties which may cause our expectations not to occur or to differ significantly from what was expected and should not be considered an offer or solicitation to buy or sell a security.

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